Consensus behavior for Hegselmann-Krause model

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Conference in honor of Shih-Hsien Yu's 60 Brithday, SNU
April 4, 2024

Figure: Taken on April 4, 2024. The fist family photo since 2014!

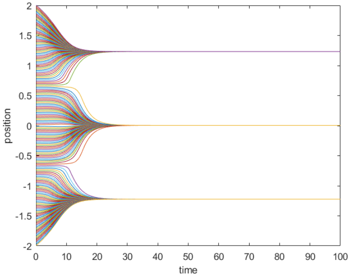
Outline

Hegselmann-Krause opinion formation model

Hegselmann-Krause model (2002') with leadership and time delay

$$
\begin{cases}
\dot{x}_0(t) = u(t, \tau_1), \\
\dot{x}_i(t) = \dfrac{1}{N} \displaystyle\sum_{j \neq i}^{N} a(|x_j(t - \tau_2) - x_i(t)|)(x_j(t - \tau_2) - x_i(t)) \\
\qquad\qquad + \gamma \phi(|x_0(t - \tau_3) - x_i(t)|)(x_0(t - \tau_3) - x_i(t)).
\end{cases}
$$

$a(\cdot)$: interactions take place in the bounded domain of confidence, e.g., $a(r) = 1$if $r \in [0, \delta]$, $a(r) = 0$ if $r \geqslant \delta + \epsilon$.



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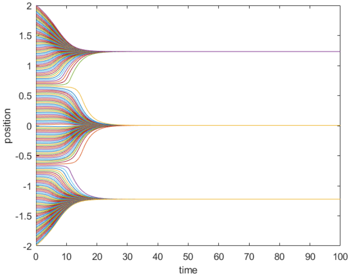

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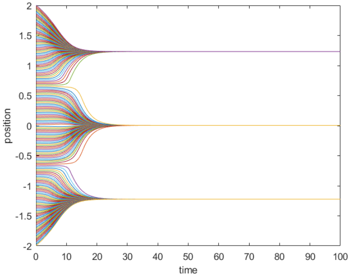

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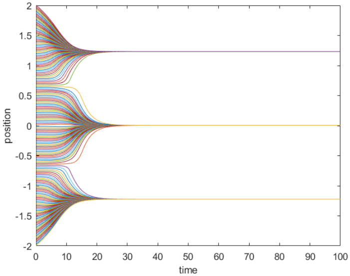

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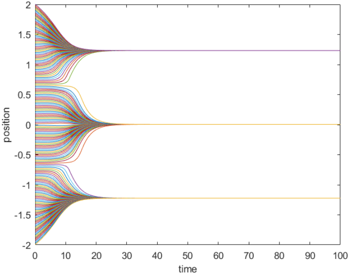

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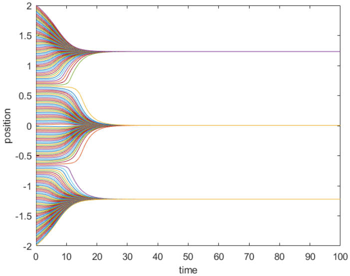

Backgrounds for the opinion dynamics

- Trace back to earlier work by Lazarfeld and Kat in 1940-50s
  French model (1956, Psychological Rev.)
  DeGroot model (1974, J Am. Stat. Assoc.)
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- Hegselmann-Krause model (02', J. Artif. Soc. Soc. Simul.)
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  S. Wongkaew, M. Caponigro & A. Borzì (15', Math. Mod. Meth. Appl. Sci.)
The control through the leadership

- Modified Hegselmann-Krause model (communication weights depend on the particle distance)
  Y. Choi, A. Paolucci & C. Pignotti (21', Math. Meth. Appl. Sci.)
  Consensus with small time delay
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  Consensus with any length of time delay

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Backgrounds for the opinion dynamics

Definition of “opinion”

- the attitude of individuals to an event/object/action
- the sets of cultural traits characterizing an individual
- an agent’s skill in a particular field
- an agents knowledge set

Outline

1. Backgrounds
2. Unconditional consensus with a special control
   - main results
   - Shrinkage estimate
3. Controllability
   - Simple control
   - Model predictive control
4. Other related works and future directions

Unconditional consensus: same delays

S. Wongkaew, M. Caponigro & A. Borzì (15', Math. Mod. Meth. Appl. Sci.)

$$\begin{cases} \dot{x}_0(t) = u(t,\tau), \\ \dot{x}_i(t) = \dfrac{1}{N}\displaystyle\sum_{j\neq i}^{N} a(|x_j(t-\tau)-x_i(t)|)(x_j(t-\tau)-x_i(t)) \\ \qquad\qquad + \gamma\phi(|x_0(t-\tau)-x_i(t)|)(x_0(t-\tau)-x_i(t)), \end{cases}$$

with

$$u(t,\tau) = \gamma\alpha(t)\sum_{j=1}^{N}\phi(x_j(t-\tau)-x_0(t))(x_j(t-\tau)-x_0(t)),$$

where

$$\alpha(t) := \frac{1}{2}\min\left\{\frac{\phi(x_\iota(t-\tau)-x_0(t))}{N-\phi(x_\iota(t-\tau)-x_0(t))}, \frac{2K}{\gamma\sum_i |x_i(t-\tau)-x_0(t)|}\right\}.$$

and $\iota=\iota(t)$ be the smallest index in $\{1,\ldots,N\}$ such that

$$|x_\iota(t-\tau)-x_0(t)| \geqslant |x_j(t-\tau)-x_0(t)|,\ j=1,\ldots,N.$$

Unconditional consensus: Numerical simulations for same delays

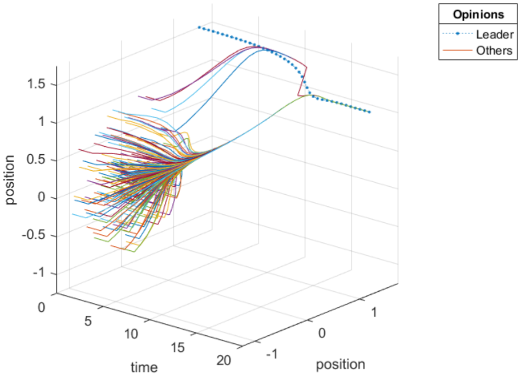

Figure: Asymptotic consensus: the two dimension positions with $\tau = 1$.

Unconditional consensus: Numerical simulations for sam delays

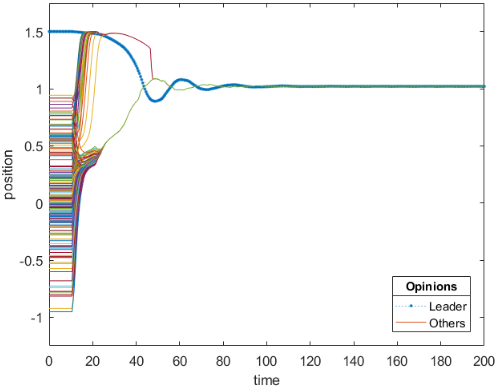

Figure: Asymptotic consensus: the first component of positions with $\tau = 10$.

Numerical simulations for consensus behavior

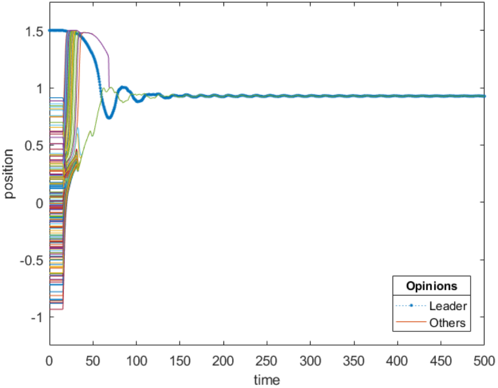

Figure: Asymptotic consensus: the first component of positions with $\tau = 15$.

Unconditional consensus result: same delays

Q: How to prove such phenomenon? i.e., the consensus result for any length of time delay.
A: Inspired by the shrinkage of group diameter estimate method proposed by J. Haskovec (21', Bull. London Math. Soc.)

Theorem

$[23' Du, Xie, Zhu]$ *The solution with any initial data reaches global asymptotic consensus through a control of leadership*

$$\lim_{t\to\infty} \|x_i(t) - x_0(t)\| = 0, \text{ for all } i \in \{1, \ldots, N\}.$$

Unconditional consensus result: Poofs for same delays

The proof consists of the following steps:

1. Rough estimate:

$$0 < m := \min_i \min_{t\in[-\tau,0]} x_i(t) \leqslant x_i(t) \leqslant \max_i \max_{t\in[-\tau,0]} x_i(t) =: M, i \in \{0,1,\ldots,N\}$$

2. Shrinkage estimate(refining the bounds):

$$m + \frac{\Gamma}{2}(M-m) \leqslant x_i(t) \leqslant M - \frac{\Gamma}{2}(M-m), t \in [3\tau, 4\tau]$$

3. Extend the shrinkage estimate to $[(4k-1)\tau, 4k\tau]$ and prove the consensus in one dimension:

$$\lim_{t\to\infty} |x_i(t) - x_0(t)| = 0, \text{ for all } i \in \{1,\ldots,N\}$$

4. Extend the result to multi-dimensions by choosing ξ as basis vectors of $\mathbb{R}^d$:

$$\begin{cases} \dot{x}_0(t)\cdot\xi = u(t,\tau)\cdot\xi, \\ \dot{x}_i(t)\cdot\xi = \dfrac{1}{N}\displaystyle\sum_{j\neq i}^{N} a_{ij}^{\tau}(t)(x_j(t-\tau)-x_i(t))\cdot\xi + \gamma\phi_{i0}^{\tau}(x_0(t-\tau)-x_i(t))\cdot\xi \end{cases}$$

Unconditional consensus result: different delays

Theorem

$[24', Du, Ha, Yuhang]$ *The solution with any initial data and different delays reaches global asymptotic consensus through a control of leadership*

$$\lim_{t\to\infty} \|x_i(t) - x_0(t)\| = 0, \text{ for all } i \in \{1, \dots, N\}.$$

- Shrinkage estimate(refining the bounds):

$$m + \frac{\Gamma}{2}(\overline{M} - m) \leqslant x_i(t) \leqslant M - \frac{\Gamma}{2}(M - \overline{m}), t \in [2\tau_b + \tau_s, 3\tau_b + \tau_s]$$

$$m := \min_{i=0,1,\dots,N} \min_{t\in[-\tau_b,0]} x_i(t), \quad M := \max_{i=0,1,\dots,N} \max_{t\in[-\tau_b,0]} x_i(t),$$
$$\overline{m} := \min_{i=0,1,\dots,N} \min_{t\in[-\tau_s,0]} x_i(t), \quad \overline{M} := \max_{i=0,1,\dots,N} \max_{t\in[-\tau_s,0]} x_i(t).$$

$\tau_b = \max\{\tau_1, \tau_2, \tau_3\}$, $\tau_s = \min\{\tau_1, \tau_2, \tau_3\}$.

Proof of step 2

Lemma

$$m + \frac{\Gamma}{2}(M - m) \leqslant x_i(t) \leqslant M - \frac{\Gamma}{2}(M - m), t \in [3\tau, 4\tau].$$

Ideas for the proof

- *Starting with refining the initial bounds using the limited speed*

$$|\dot{x}_i| \leqslant (1+\gamma)M \text{ for all } t > 0 \text{ and } i \in \{0, 1, \ldots, N\}.$$

Suppose $x_L(s_1) = m$ *and* $x_R(s_2) = M$*, where* $s_1, s_2 \in [-\tau, 0]$ *and* $L, R \in \{0, 1, \ldots, N\}$.

- *Continuing the proof by refining the bounds in some time intervals*

  Case 1: $0 \notin \{L, R\}$.

  Case 2: $0 \in \{L, R\}$.

Glimpse of the estimate for leader $x_0(t)$

Since $x_L(s_1) = m$ for $s_1 \in [-\tau, 0]$, there exists an interval $[\beta_L, \omega_L] \subseteq [-\tau, 0]$ such that $x_L(t)$ satisfies

$$m \leqslant x_L(t) \leqslant \frac{m+M}{2}, \quad t \in [\beta_L, \omega_L].$$

When $t \in [\beta_L + \tau, \omega_L + \tau]$, we have

$$\begin{aligned} \dot{x}_0(t) &= \gamma\alpha(t) \sum_{j=1}^{N} \phi_{0j}^{\tau}(x_j(t-\tau) - x_0(t)) \\ &= \gamma\alpha(t) \sum_{j=1, j\neq L}^{N} \phi_{0j}^{\tau}(x_j(t-\tau) - x_0(t)) + \gamma\alpha(t)\phi_{0L}^{\tau}(x_L(t-\tau) - x_0(t)) \end{aligned}$$

Then move the estimate of $x_0(t)$ to the time interval $[\omega_L + \tau, 4\tau]$.

Glimpse of the estimate for followers $x_i(t)$

When $t \in [2\tau, 4\tau]$,

$$\begin{aligned}\dot{x}_i(t) &= \frac{1}{N}\sum_{j\neq i} a_{ij}^{\tau}(t)(x_j(t-\tau) - x_i(t)) + \gamma\phi_{i0}^{\tau}(x_0(t-\tau) - x_i(t)) \\ &\geqslant m - x_i(t) + \gamma\phi_{i0}^{\tau}((1+\theta_{1+}e^{-4\gamma\tau})m - x_i(t)).\end{aligned}$$

Proof of step 2

Case 1: $0 \notin \{L, R\}$.

$\triangleright$ *Estimate the upper bound for the leader* $x_0(t)$ *on* $[\omega_L + \tau, 4\tau]$ *and the lower bound on* $[\omega_R + \tau, 4\tau]$*:*

$$x_0(t) \leqslant (1 - \theta_{1-} e^{-4\gamma\tau}) M, \quad t \in [\omega_L + \tau, 4\tau];$$

$$x_0(t) \geqslant (1 + \theta_{1+} e^{-4\gamma\tau}) m, \quad t \in [\omega_R + \tau, 4\tau].$$

Here

$$\theta_{1-} := \frac{1}{2\gamma} \underline{\psi} (1 - e^{-\gamma\sigma}) \left(1 - \frac{m}{M}\right), \quad \theta_{1+} := \frac{1}{2\gamma} \underline{\psi} (1 - e^{-\gamma\sigma}) \left(\frac{M}{m} - 1\right).$$

Proof of step 2

▷ *Estimate the upper and lower bounds for the followers* $\{x_i\}_{i\in\{1,\ldots,N\}}$ *on* $[3\tau, 4\tau]$:

$$x_i(t) \leqslant \left(1 - \frac{\underline{\psi}}{\underline{\psi}+1}\theta_{1-}e^{-4\gamma\tau}(1-e^{-(\underline{\psi}+1)\tau})\right)M, \quad t\in[3\tau,4\tau];$$

$$x_i(t) \geqslant \left(1 + \frac{\underline{\psi}}{\underline{\psi}+1}\theta_{1+}e^{-4\gamma\tau}(1-e^{-(\underline{\psi}+1)\tau})\right)m, \quad t\in[3\tau,4\tau].$$

Proof of step 2

Summarization $\{x_i\}_{i\in\{0,\dots,N\}}$ in $t\in[3\tau,4\tau]$

Case 1: $0\notin\{L,R\}$

$$\left(1+\frac{\psi}{\underline{\psi}+1}\theta_{1+}e^{-4\gamma\tau}(1-e^{-(\underline{\psi}+1)\tau})\right)m\leqslant x_i(t)\leqslant\left(1-\frac{\psi}{\underline{\psi}+1}\theta_{1-}e^{-4\gamma\tau}(1-e^{-(\underline{\psi}+1)\tau})\right)M.$$

Case 2: $0\in\{L,R\}$

$$\left(1+\frac{\psi}{\underline{\psi}+1}\theta_{1+}e^{-4\gamma\tau}(1-e^{-(\underline{\psi}+1)\tau})\right)m\leqslant x_i(t)\leqslant(1-\theta_{2-}e^{-4(\gamma+1)\tau}(1-e^{-\underline{\psi}\tau}))M.$$

Conclusion

$$m+\frac{\Gamma}{2}(M-m)\leqslant x_i(t)\leqslant M-\frac{\Gamma}{2}(M-m),\quad t\in[3\tau,4\tau],\quad i\in\{0,1,\dots,N\}.$$

Here $\Gamma\in(0,1)$ is defined by

$$\Gamma:=e^{-4\gamma\tau}\underline{\psi}\eta,$$

where,

$$\eta:=\min\left\{\frac{\psi}{\gamma(\underline{\psi}+1)}(1-e^{-\gamma\sigma})(1-e^{-(\underline{\psi}+1)\tau}),\ \frac{1}{\gamma+1}(1-e^{-(\gamma+1)\sigma})(1-e^{-\underline{\psi}\tau})e^{-4\tau}\right\}.$$

Outline

Simple control to any target position: theoretical result

$$\begin{cases} \dot{x}_0(t) = u(t), \\ \dot{x}_i(t) = \dfrac{1}{N}\displaystyle\sum_{j\neq i}^{N} a(|x_j(t-\tau)-x_i(t)|)(x_j(t-\tau)-x_i(t)) \\ \qquad + \gamma\phi(|x_0(t-\tau)-x_i(t)|)(x_0(t-\tau)-x_i(t)). \end{cases}$$

Lemma

$[23' Du, Xie, Zhu]$ *Let* $\{x_i\}_{i\in\{0,1,\ldots,N\}}$ *be the solution of above system with* $u(t)=0$ *for any* $t \geqslant 0$, $x_{0,0}(s) = x_0$ *for any* $s \in [-\tau, 0]$. *Then all agents reach global asymptotic consensus under the leader's control*

$$\lim_{t\to\infty} |x_i(t) - x_0| = 0, \text{ for all } i \in \{1, \ldots, N\}.$$

Theorem

$[23' Du, Xie, Zhu]$ *For any given* $\bar{x} \in \mathbb{R}^d$, *there exists an admissible control* $u(\cdot) : [0, +\infty) \to \mathbb{R}^d$ *for system with same delays, which steers all agents to* $\bar{x}$, *namely*

$$\lim_{t\to\infty} x_i(t) = \bar{x}, i \in \{0, 1, \ldots, N\}.$$

Simple control to any target position

Proof.

Step 1. Mild control period Take the control

$$u(t) = \gamma\alpha(t)\sum_{j=1}^{N}\phi_{0j}^{\tau}(x_j(t-\tau) - x_0(t)).$$

for any given $\varepsilon > 0$, there exists a positive time $t_0 > 0$ and position x^* such that

$$\max_{i\in\{0,1,\dots,N\}}\max_{t\in[t_0-\tau,t_0]}|x_i(t) - x^*| \leqslant \varepsilon.$$

Step 2. Strong control period For $t > t_0$, take the control

$$u(t) = K\frac{\bar{x} - x_0(t)}{|\bar{x} - x_0(t)|}.$$

There exists another time $t_1 > t_0$ such that $x_0(t_1) = \bar{x}$.
Step 3. Free control period For $t > t_1$, consider $u(t) = 0$. According to the above lemma, all agents reach global asymptotic consensus. □

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Simple control to any target position: numerical simulation

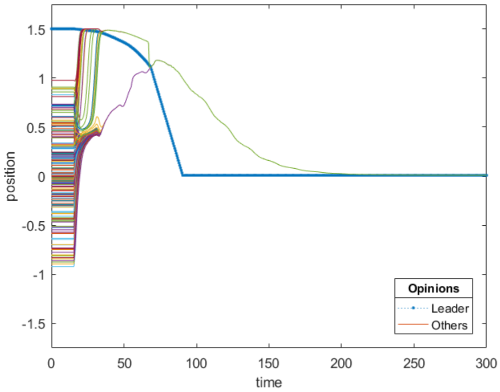

Figure: Simple control strategy: the first component of positions with $\tau = 15$.

Simple control to any target position: numerical simulation

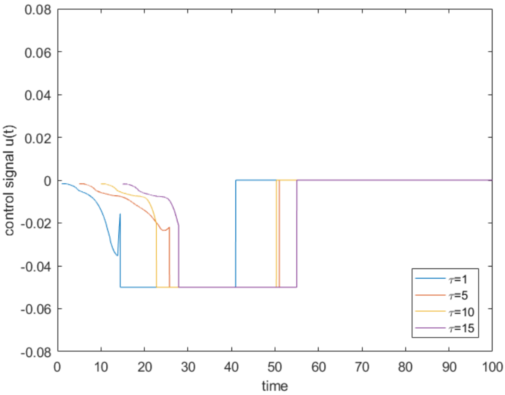

Figure: Time series diagram of control signal with $\tau = 1, 5, 10, 15$.

MPC Algorithm

Design a closed-loop control strategy for the delayed model in order to track a given sequence of desired positions in time:

$$\begin{cases} \dot{x}_0(t) = u(t), \\ \dot{x}_i(t) = \dfrac{1}{N} \displaystyle\sum_{j \neq i}^{N} a(|x_j(t-\tau_2) - x_i(t)|)(x_j(t-\tau_2) - x_i(t)) \\ \qquad\qquad + \gamma\phi(|x_0(t-\tau_3) - x_i(t)|)(x_0(t-\tau_3) - x_i(t)), \end{cases} \tag{1}$$

The optimal control problem can be written in the following form

$$\begin{aligned} \min \quad & J(x_0, u) = \frac{1}{2}\left|x_0(t_f) - x_{\mathrm{des}}(t_f)\right|^2 + \frac{\eta}{2}\int_{t_0}^{t_f} \left|x_0(t) - x_{des}(t)\right|^2 dt \\ & \qquad + \frac{\nu}{2}\int_{t_0}^{t_f} |u(t)|^2 dt, \\ s.t. \quad & \dot{x}_0(t) = u(t), \quad x_0(t_0) = x_{0,0}, \end{aligned} \tag{2}$$

where $x_0 \in H^1\left((t_0, t_f); \mathbb{R}\right)$. The control function $u \in L^2\left((t_0, t_f); \mathbb{R}\right)$.

MPC Algorithm

The associated Lagrange multiplier $\lambda(t)$ such that (x_0^*, λ^*, u^*) satisfies the follow system,

$$\begin{aligned}
&\dot{x_0}(t) = u(t), \quad x_0(t_0) = x_0, \\
&\dot{\lambda}(t) = -\nabla_{x_0} \mathbf{H}(t), \quad \lambda(t_f) = x_0(t_f) - x_{des}(t_f), \\
&\nu u(t) + \lambda(t) = 0, \quad t \in [t_0, t_f].
\end{aligned} \tag{3}$$

The first equation of (3) is the state equation, the second equation is the adjoint equation with the terminal condition, and the third equation is the optimality condition equation. The adjoint equation is written in detail as follows:

$$\dot{\lambda}(t) = -\nabla_{x_0} \mathbf{H}(t) = -\eta \left(x_0(t) - x_{des}(t)\right). \tag{4}$$

MPC Algorithm

Algorithm 1 (MPC control) Set $k = 0$:
(1) Assign the initial condition, $\hat{\mathbf{x}}(t_{kd1}) = \hat{\mathbf{x}}_{kd1}$, $x_0(t_{kd2}) = x_{0,kd2}$, $\mathbf{x}(t_k) = \mathbf{x}_k$ and the target $x_{\text{des}}(t_{k+1})$;
(2) In $(t_k, t_k + 1)$, solve (2), thus obtain the optimal pair (x_0, u);
(3) Then, solve (1) and get the value of $x_i(t)$ in $(t_k, t_k + 1)$, $i = 1, \dots, N$;
(4) If $t_{k+1} < T$, set $k := k + 1, \mathbf{x}_k = \mathbf{x}(t_k)$, go to 1;
(5) End.

Algorithm 2 (Evaluation of the gradient at u)
(1) Solve the discrete model in the optimality system with the given initial conditions;
(2) Solve the adjoint equation in (4) with the computed terminal condition;
(3) Compute the gradient $\nabla_u \hat{J}(u)$ using optimality condition;
(4) End.

Algorithm 3 (Gradient descent method with BB update) Require: $tol > 0$, $k = 0$, $k_{\max}$, u^0, u^{-1}.
While $\left\|\nabla J\left(u^k\right)\right\| > tol$ and $k < k_{\max}$ do
(1) Obtain x_0^k from (1) with u^{-1};
(2) Obtain λ^k from (4) with x_0^k;
(3) Evaluate the gradient $\nabla J\left(u^k\right)$;
(4) Compute the step

$$\alpha_k := \frac{\left\langle u^k - u^{k-1}, \nabla J\left(u^k\right) - \nabla J\left(u^{k-1}\right)\right\rangle}{\left\|\nabla J\left(u^k\right) - \nabla J\left(u^{k-1}\right)\right\|^2}; \tag{5}$$

(5) Update $u^{k+1} = u^k - \alpha_k \nabla J\left(u^k\right)$;
(6) Set $k := k + 1$;
End while.

Outline

Other related works and future directions

• Cucker-Smale model with leadership and time delay (24', Cheng, Du and Sang) track a given sequence of desired positions in time

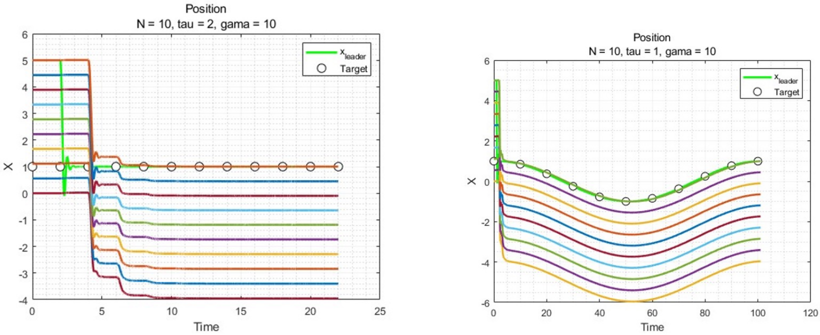

Figure: MPC for the delayed Cucker-Smale system with leadership.

• Phase-locking for Kuramoto model with pacemaker and delay: remove the positive initial order parameter condition $R(0) > 0$

• Time delay of the process type (23' Du Han and Wang, 24', Du, Dong and Yin)

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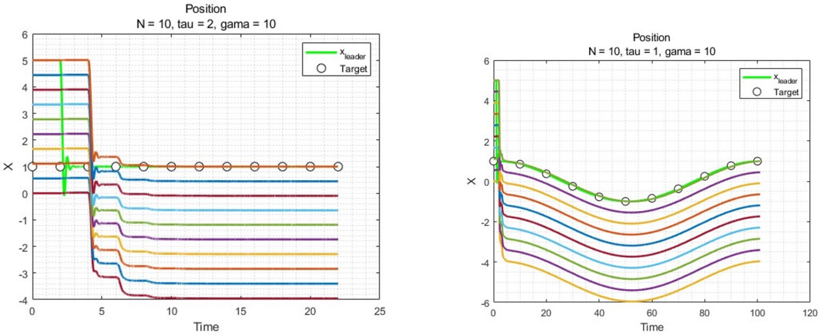

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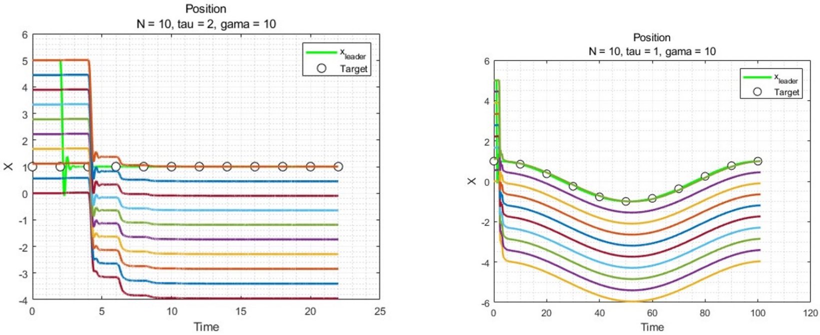

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Happy Birthday to Prof. Shih-Hsien Yu!